

Phase 11d: Lambda Expressions

Lambda expressions provide a concise way to define anonymous functions.

(a)

Simple lambda expression

$$(\lambda x : \mathbb{N} \bullet x^2)$$

(b)

Multi-variable lambda

$$(\lambda x, y : \mathbb{N} \bullet x \wedge y)$$

(c)

Lambda with comparison

$$(\lambda x : \mathbb{N} \bullet x > 0)$$

(d)

Nested lambda expression

$$(\lambda x : X \bullet (\lambda y : Y \bullet x \wedge y))$$

(e)

Lambda with quantifier in body

$$(\lambda x : X \bullet \forall y : Y \bullet x \vee y)$$